BASE CHANGE FOR THE PUSHFORWARD ALONG A FINITE MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the equality, with the reduction and the affine computation both inside the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Finite morphism of schemes and its stability under base change. Same chapter together with the chapter Morphisms, the definition of a finite morphism and the fact that the base change f' is again finite. Supplies the hypothesis and the affineness used in the reduction.
- Pushforward along a finite morphism computed by sections over the base change. Same chapter, the description of f_*F for finite f by the sections over the base change, and the comparison of the two sides in the affine case. Supplies the computation of both sides in the affine case.

Source reference. Original lemma 06UM: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8308–8308. Navigation pointer invoked at proof line 8308; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original theorem 03PU: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 3546–3546. Navigation pointer invoked at proof line 8286; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original proposition 03QP: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8245–8245. Navigation pointer invoked at proof line 8330; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 03Q1: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 4649–4649. Navigation pointer invoked at proof line 8332; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Consider a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

of schemes with f a finite morphism. For any sheaf of sets $\mathcal{F}$ on $X_{étale}$ we have $f'_*(g')^{-1}\mathcal{F} = g^{-1}f_*\mathcal{F}$.

Proof. In great generality there is a pullback map $g^{-1}f_*\mathcal{F} \to f'_*(g')^{-1}\mathcal{F}$, see Sites, Section 06UM. It suffices to check on stalks (Theorem 03PU). Let $\overline{y}' : \mathrm{Spec}(k) \to Y'$ be a geometric point. We have

$$\begin{aligned} (f'_*(g')^{-1}\mathcal{F})_{\overline{y}'} &= \prod\nolimits_{\overline{x}' : \mathrm{Spec}(k) \to X',\ f' \circ \overline{x}' = \overline{y}'} ((g')^{-1}\mathcal{F})_{\overline{x}'} \\ &= \prod\nolimits_{\overline{x}' : \mathrm{Spec}(k) \to X',\ f' \circ \overline{x}' = \overline{y}'} \mathcal{F}_{g' \circ \overline{x}'} \\ &= \prod\nolimits_{\overline{x} : \mathrm{Spec}(k) \to X,\ f \circ \overline{x} = g \circ \overline{y}'} \mathcal{F}_{\overline{x}} \\ &= (f_*\mathcal{F})_{g \circ \overline{y}'} \\ &= (g^{-1}f_*\mathcal{F})_{\overline{y}'} \end{aligned}$$

The first equality by Proposition 03QP. The second equality by Lemma 03Q1. The third equality holds because the diagram is a cartesian square and hence the map

$$\{\overline{x}' : \mathrm{Spec}(k) \to X',\ f' \circ \overline{x}' = \overline{y}'\} \longrightarrow \{\overline{x} : \mathrm{Spec}(k) \to X,\ f \circ \overline{x} = g \circ \overline{y}'\}$$

sending $\overline{x}'$ to $g' \circ \overline{x}'$ is a bijection. The fourth equality by Proposition 03QP. The fifth equality by Lemma 03Q1. □